

BEYOND THE FORD-CIRCLE LAW: PARABOLIC RENORMALIZATION AND THE SHAPE OF MANDELBROT SATELLITE BULBS (ABRIDGED VERSION)

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ABSTRACT. To first order in $1/q$, the satellite of the Mandelbrot set at internal angle p/q is a round disc of diameter $2 \sin(\pi p/q) q^{-2}$; this is the Ford-circle law. We study the first correction. In the bulb-scale variable $u = q^2 \varepsilon$, the first coefficient of the cycle multiplier that depends on p/q is $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$, where $\iota_{p/q}$ is the holomorphic index of the parabolic point; we prove this, identify $\kappa + \frac{1}{2}$ with the index of a rotation quotient, and obtain an $O(q^2)$ recursion. Numerically, the correction is governed by parabolic renormalization. The limit of $\kappa(1/q)$ is the horn-map index $\kappa_0 = 0.0238258874 \dots - 0.0523046591 \dots i$ of $z + z^2$, to 24 digits. Along $q = pN + r$ the limits of κ and of the bulb diameter follow the horn germ times the Inou–Shishikura phase, and an exact Möbius form of that phase accounts for the whole $O(1/q)$ correction. Iterating along large partial quotients gives constants that converge geometrically (ratio 0.042) to a constant κ_* of renormalization composed with complex conjugation. At rationals, κ has distinct one-sided limits, given by the upper and lower horn maps of $e^{2\pi i p/q} z + z^2$; we verify this numerically at $\frac{1}{3}, \frac{1}{4}, \frac{2}{5}, \frac{3}{7}$ and conjecture it at every rational. Satellites of other components follow the parabolic germs at their roots.

1. INTRODUCTION

Let $c(\lambda) = \lambda/2 - \lambda^2/4$ parametrize the main cardioid of the Mandelbrot set by the multiplier $\lambda \in \mathbb{D}$ of its attracting fixed point; equivalently we use $f_\lambda(z) = \lambda z + z^2$. At $\lambda_0 = e^{2\pi i p/q}$ the satellite $W_{p/q}$ of period q is attached [3]. Its size is of order q^{-2} [7], and to first order it is a disc of diameter $2|c'(\lambda_0)|q^{-2} = 2 \sin(\pi p/q) q^{-2}$ [5]. Up to the factor $\sin(\pi p/q)$ these are the Ford circles [6]. We study the first correction to this law.

Put $\lambda = \lambda_0 e^\varepsilon$, let $\rho_{p/q}(\varepsilon)$ be the multiplier of the q -cycle that bifurcates from 0, and write

$$(1) \quad R_{p/q}(u) := \rho_{p/q}(u/q^2) = \sum_{k \geq 0} r_k(p/q) u^k.$$

Up to a conformal change of variable, $W_{p/q} = \{|R_{p/q}| < 1\}$. We show that $r_0 = 1$ and $r_1 = -1$ for all p/q , so that to first order $W_{p/q}$ is the disc $|u - 1| < 1$; this is the Ford law. The first coefficient that depends on p/q is $\kappa(p/q) := r_2(p/q)$, and it governs the leading deviation of $W_{p/q}$ from a disc. The bulb size is measured by $G(p/q) = \frac{1}{2}|u_a|(1 + O(q^{-2}))$, where $R_{p/q}(u_a) = -1$ and $G = 1$ to first order.

Results. Section 2 contains the proved statements: the index formula for κ , and a normal form that computes it in $O(q^2)$ operations. The remaining sections are numerical, and each claim is stated as a conjecture together with its evidence. We find that, up to $O(q^{-2})$, the bulb

Date: September 2026.

2020 Mathematics Subject Classification. Primary 37F46; Secondary 37F25, 37F10.

Key words and phrases. Mandelbrot set, satellite components, Ford circles, holomorphic index, horn map, parabolic renormalization.

is described by the near-parabolic renormalization of f_{λ_0} in the sense of Inou and Shishikura [8], composed with an exact phase. This picture yields five constants, collected in Table 1. The full version [1] contains proofs omitted here, degree-three details, error analysis and data; all code and data are in [12].

2. EXACT RESULTS

Write $f = f_{\lambda_0}$. The fixed point 0 has a single cycle of q petals, so $f^q(z) = z + az^{q+1} + O(z^{q+2})$ with $a \neq 0$. Let $\iota_{p/q} = \text{Res}_{z=0} dz/(z - f^q(z))$ be its holomorphic index [11].

Theorem 2.1 (Index formula). $\rho_{p/q}(\varepsilon) = 1 - q^2\varepsilon + q^3(\iota_{p/q} - \frac{1}{2})\varepsilon^2 + O(\varepsilon^3)$. Equivalently, $r_0 = 1$, $r_1 = -1$ and $\kappa(p/q) = (\iota_{p/q} - \frac{1}{2})/q$.

Sketch. For small $\varepsilon \neq 0$, f_λ^q has $q + 1$ simple fixed points near 0: the point 0, with multiplier $e^{q\varepsilon}$, and one q -cycle, with multiplier ρ . The sum of their indices converges to $\iota_{p/q}$, which gives

$$\frac{1}{1 - e^{q\varepsilon}} + \frac{q}{1 - \rho(\varepsilon)} \longrightarrow \iota_{p/q}.$$

The first term is $-1/(q\varepsilon) + \frac{1}{2} + O(\varepsilon)$. Boundedness of the sum forces $1 - \rho = q^2\varepsilon + O(\varepsilon^2)$, and the constant term gives the coefficient of ε^2 . $\square$

The proof uses only the multiplicity $q + 1$, so it applies to the satellites of every hyperbolic component of $z^d + c$, with the first-return map in place of f . Since f^q has coefficients in $\mathbb{Z}[\lambda_0]$, $\kappa(p/q) \in \mathbb{Q}(\zeta_q)$, and the values $\kappa(p'/q)$ form a Galois orbit; for instance $\kappa(\frac{1}{2}) = -\frac{3}{16}$ and $\kappa(\frac{1}{3}) = (-241 - 16\sqrt{3}i)/2646$.

Theorem 2.2 (Normal form). Let $\mathcal{N}(z) = \lambda_0 z(1 + b_1 z^q + b_2 z^{2q})$ be the resonant normal form of f to order $2q + 1$. Its coefficients satisfy an explicit triangular recursion of cost $O(q^2)$, and

$$a = qb_1, \quad \kappa(p/q) = \frac{1}{2} - \frac{1}{2q} - \frac{1}{2q^2} + \frac{b_2}{(qb_1)^2} = \iota(Q_{p/q}) - \frac{1}{2},$$

where $Q_{p/q}(W) = \mathcal{N}^q(W^{1/q})^q$ is the rotation quotient of $\mathcal{N}^q$.

Composing f symbolically q times costs $O(q^3)$ and suffers severe cancellation, since $|a(1/q)|$ grows like $(q/2\pi)^q$. The recursion is stable; in Arb ball arithmetic [2] it certifies $\kappa(1/q)$ for $q \leq 32768$.

3. THE HORN MAP OF $z + z^2$ AND κ_0

Let $E(Z) = \Phi_{\text{att}} \circ \Psi_{\text{rep}}(Z) = Z + \sum_{n \geq 1} a_n e^{2\pi i n Z}$ be the upper horn map of $f_1(z) = z + z^2$, with the standard normalization $a_0 = 0$ [10, 13]. Let

$$\mathcal{P}_0(W) = W \exp\left(2\pi i \sum_{n \geq 1} a_n W^n\right)$$

be its parabolic renormalization [8, 9].

Lemma 3.1. $\iota(\mathcal{P}_0) - \frac{1}{2} = a_2/(2\pi i a_1^2)$, independently of the normalization of the Fatou coordinates.

Proof. $\mathcal{P}_0(W) = W + AW^2 + BW^3 + \dots$ with $A = 2\pi i a_1$ and $B = 2\pi i a_2 + \frac{1}{2}A^2$, and $\iota = B/A^2$. A translation of both Fatou coordinates by τ sends $a_n \mapsto a_n e^{2\pi i n \tau}$, which leaves a_2/a_1^2 unchanged. $\square$

constant	germ	value	bulbs	
κ_0	$z + z^2$	$0.02382588740220057 - 0.05230465911400367 i$	$3 \cdot 10^{-28}$	§3
$\kappa_0^{(3)}$	$v + v^2 + \frac{1}{3}v^3$	$0.1435992780952916 - 0.03031164789850254 i$	$1.3 \cdot 10^{-6}$ ($q = 1024$)	§7
$\kappa_{1/2}^+$	$-z + z^2$	$0.01840526161796478 - 0.04290372388023048 i$	$3 \cdot 10^{-10}$	§6
κ_*	$\mathcal{G}_* = \mathcal{P}(\bar{\mathcal{G}}_*)$	$0.011929183413 - 0.037455493752 i$	levels ≤ 3	§5
G_0	$e^u \mathcal{P}_0$	1.02641539	$4 \cdot 10^{-8}$	§4

TABLE 1. The constants. “Bulbs” is the difference between the horn-map value and the best extrapolated bulb limit ($\kappa_0^{(3)}$: error at $q = 1024$, decaying like q^{-2}). The extrapolation uncertainties are about $3 \cdot 10^{-25}$ for κ_0 , $2 \cdot 10^{-7}$ for $\kappa_{1/2}^+$ and 10^{-6} for G_0 .

Numerical result 3.2. The horn map, computed from exact Fatou series at 60 and 80 digits, gives

$$\begin{aligned} \operatorname{Re} \kappa_0 &= 0.023825887402200569000729189645402, \\ \operatorname{Im} \kappa_0 &= -0.052304659114003666316522486207571, \end{aligned}$$

where $\kappa_0 := \iota(\mathcal{P}_0) - \frac{1}{2}$. Least-squares fits $\kappa(1/q) = \sum_{j \leq 10} \gamma_j q^{-j}$ over certified values $256 \leq q \leq 32768$ give γ_0 equal to this value to $3 \cdot 10^{-28}$, and $\gamma_1 = -i/(2\pi)$ to $2 \cdot 10^{-23}$. Other stable fits move γ_0 by at most $3 \cdot 10^{-25}$.

Conjecture 3.3. $\kappa(1/q) = \kappa_0 - \frac{i}{2\pi q} + O(q^{-2})$.

4. THE RENORMALIZATION PRINCIPLE

Near-parabolic renormalization $\mathcal{R}$ [8] assigns to $f_{e^{2\pi i \alpha}}$ a germ with multiplier $e^{-2\pi i/\alpha}$, and parabolic implosion [10, 13] identifies its limits with $\mathcal{P}_0$ times a phase. The phase is exact.

Proposition 4.1 (Kinematics). *If $e^{2\pi i \alpha} = \lambda_0 e^{u/q^2}$, then $e^{-2\pi i/\alpha} = e^{-2\pi i q/p} e^{\tilde{u}/p^2}$ with $\tilde{u} = u/(1 + u/(2\pi i p q))$.*

Proof. $\alpha = (2\pi i p q + u)/(2\pi i q^2)$, and $2\pi i p^2(q/p - 1/\alpha) = \tilde{u}$. $\square$

Fix p and r coprime, let $q = pN + r$, $\mu = e^{-2\pi i r/p}$, and let $\rho_{p,r}(u)$ be the multiplier of the p -cycle of $\mathcal{M}_u = \mu e^{u/p^2} \mathcal{P}_0$ that bifurcates from 0.

Conjecture 4.2 (Renormalization principle). $R_{p/q}(u) = \rho_{p,r}(\tilde{u}(u)) + O(q^{-2})$ coefficientwise, and uniformly on a neighbourhood of the arc from 0 to the point where $\rho_{p,r} = -1$.

Comparing coefficients gives $\kappa(p/q) = \kappa_{p,r} - i/(2\pi p q) + O(q^{-2})$, where

$$(2) \quad \kappa_{p,r} = \frac{\iota((\mu \mathcal{P}_0)^p) - \frac{1}{2}}{p},$$

by Theorem 2.1 applied to $\mu e^{\varepsilon} \mathcal{P}_0$. The first-order term involves no property of the germ.

Evidence. In all nine classes with $2 \leq p \leq 5$, (2) agrees with the extrapolated bulb limits to between $6 \cdot 10^{-9}$ and $2.6 \cdot 10^{-8}$, within the extrapolation uncertainty. The opposite phase fails.

The predicted bulb sizes agree with the bulbs to $9 \cdot 10^{-7}$ in all ten classes with $p \leq 5$. For $p = 1$ this gives diameters $2\pi G_0 q^{-3}(1 + O(q^{-1}))$ for the $1/q$ -bulbs at the cusp, with $G_0 = 1.02641539 \dots$

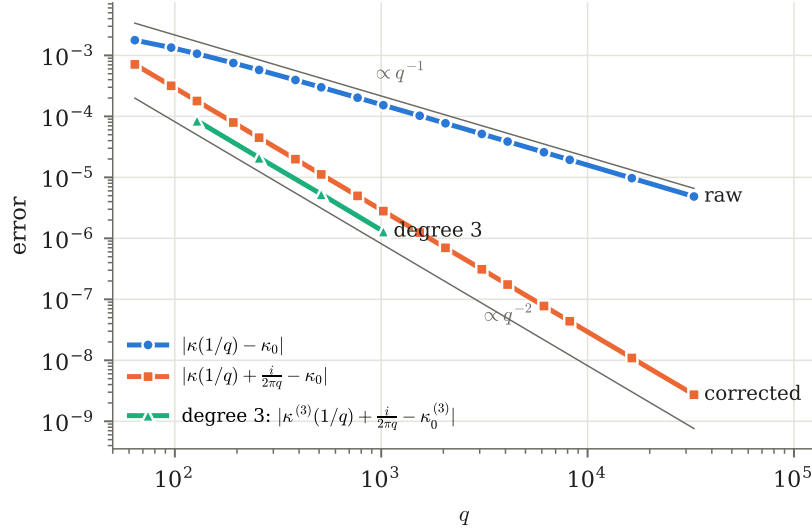


FIGURE 1. $\kappa(1/q) \rightarrow \kappa_0$ (degree two) and $\kappa^{(3)}(1/q) \rightarrow \kappa_0^{(3)}$ (degree three). After the universal correction $-i/(2\pi q)$ the error decays like q^{-2} .

For $2 \leq k \leq 8$, $(p, r) \in \{(1, 1), (2, 1), (3, 1), (3, 2)\}$ and $N = 64, \dots, 1024$, the residuals $|r_k(p/q) - [u^k]_{\rho_{p,r}}(\tilde{u})|$ fall by a factor 3.8–4.3 per doubling of N , that is, like q^{-2} . Without $u \mapsto \tilde{u}$ the factor is only 1.1–2.0 (Figure 2). Numerically, then, the entire $O(1/q)$ part of $R_{p/q}$ is kinematic.

5. THE HIERARCHY AND κ_*

The renormalization $\mathcal{R}$ acts on continued fractions as the Gauss map, after conjugation by $z \mapsto \bar{z}$. Therefore, when the first k partial quotients are large, each renormalization should be close to $\mathcal{P}$ applied to the complex conjugate of the previous one. Normalize germs to $v + v^2 + O(v^3)$, write $\bar{\mathcal{G}}(W) = \overline{\mathcal{G}(\bar{W})}$, and put

$$\mathcal{G}_1 = \mathcal{P}(f_1), \quad \mathcal{G}_{k+1} = \mathcal{P}(\bar{\mathcal{G}}_k), \quad \kappa_k = \iota(\mathcal{G}_k) - \frac{1}{2}.$$

Numerical result 5.1. Two independent settings agree to $2 \cdot 10^{-14}$ for $k \leq 7$. The differences $\kappa_{k+1} - \kappa_k$ contract antilinearly, $\Delta \mapsto \nu \bar{\Delta}$ with $|\nu| = 0.0419$. This is the signature of the antilinear derivative of $\mathcal{G} \mapsto \mathcal{P}(\bar{\mathcal{G}})$. Extrapolation gives

$$\kappa_* = \lim_k \kappa_k = 0.011929183413 - 0.037455493752 i.$$

Without the conjugation our code reproduces the fixed point of Lanford and Yampolsky [9]: cubic coefficient $0.5142768 - 0.0346276 i$, eigenvalue ratio $-0.01764 + 0.04038 i$. That operator, however, does not govern the bulbs.

Conjecture 5.2. $\lim_{N_k \rightarrow \infty} \dots \lim_{N_1 \rightarrow \infty} \kappa([0; N_1, \dots, N_k])$ equals κ_k for k odd and $\overline{\kappa_k}$ for k even.

This holds numerically for $k \leq 3$. At $k = 2$ the bounded- p route gives $\overline{\kappa_2}$ to $1.2 \cdot 10^{-8}$, and the diagonal $[0; N, N]$ gives it to $3 \cdot 10^{-10}$. At $k = 3$ the diagonal $[0; N, N, N]$, with $q \leq 64080$, gives κ_3 to 10^{-5} , while $|\kappa_3 - \kappa_2| = 7.8 \cdot 10^{-4}$.

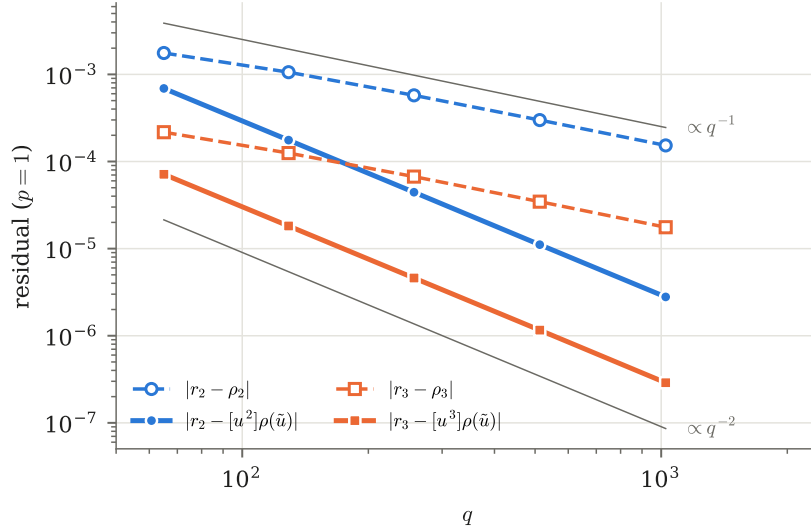


FIGURE 2. Residuals of r_2, r_3 of $R_{1/q}$ against the renormalized family, with (solid) and without (dashed) the Möbius substitution $u \mapsto \tilde{u}$.

6. LIMITS AT PARABOLIC ROOTS

Let $p/q = [0; a_1, \dots, a_\ell]$, using either of its two continued-fraction expansions. Let τ be a fixed tail with $[0; \tau] = r_\tau/p_\tau$, where $r_\tau = 0$ and $p_\tau = 1$ for $\tau = \emptyset$. As $N \rightarrow \infty$ the satellites $[0; a_1, \dots, a_\ell, N, \tau]$ approach the root of $W_{p/q}$, from above if ℓ is even and from below if ℓ is odd. There the cusp germ is replaced by the q -petal germ $f_{e^{2\pi i p/q}}(z) = e^{2\pi i p/q} z + z^2$.

The construction of Section 3 extends to $g = f_{e^{2\pi i p/q}}^q$. There is a formal Fatou coordinate

$$\Phi = -z^{-q}/(qa) + \sum s_k z^k + \beta \log z,$$

where $\beta = \text{résit}(g) = \frac{q+1}{2} - \iota(g)$; this follows from a residue computation. The orbits from a repelling petal above or below an escape band land in different attracting petals, and are carried back by f^m with $m \equiv -jp^{-1} \pmod{q}$. This defines an upper and a lower horn map, with germs $\mathcal{P}_{p/q}^\pm$ normalized tangent to the identity. Put $\kappa_{p/q}^\pm = \iota(\mathcal{P}_{p/q}^\pm) - \frac{1}{2}$. Then $\kappa_{0/1}^\pm = \kappa_0, \overline{\kappa_0}$ and $\kappa_{p/q}^- = \overline{\kappa_{(q-p)/q}^+}$, but for $q \geq 3$ the constants $\kappa_{p/q}^+$ and $\kappa_{p/q}^-$ at the same p/q are not conjugate.

Conjecture 6.1 (Root limits). Let $\mu_\tau = e^{-2\pi i r_\tau/p_\tau}$. From above,

$$\lim_{N \rightarrow \infty} \kappa([0; a_1, \dots, a_\ell, N, \tau]) = \frac{\iota((\mu_\tau \mathcal{P}_{p/q}^+)^{p_\tau}) - \frac{1}{2}}{p_\tau}.$$

From below the same formula holds with $\overline{\mu_\tau} \mathcal{P}_{p/q}^-$ in place of $\mu_\tau \mathcal{P}_{p/q}^+$. In particular κ has the distinct one-sided limits $\kappa_{p/q}^\pm$ at every rational.

The conjugate phase from below is forced by $\kappa(1-x) = \overline{\kappa(x)}$. Table 2 tests the conjecture in fifteen cases. Eight have $\tau = \emptyset$, at $\frac{1}{3}, \frac{1}{4}, \frac{2}{5}, \frac{3}{7}$ from both sides. Seven have tails with $p_\tau = 2, 3$, at $\frac{1}{3}, \frac{1}{2}$ and $\frac{1}{4}$; four of these approach from below with a non-real phase. All agree to $6 \cdot 10^{-11}$. At $\frac{1}{3}$, for instance, $\kappa_{1/3}^- = 0.0158408 + 0.0401752i$ and $\kappa_{1/3}^+ = 0.0165812 - 0.0395047i$.

p/q	side	τ	horn map of $e^{2\pi i p/q} z + z^2$	bulbs, extrapolated	$ \Delta $
1/3	below	$\emptyset$	$0.015840773234 + 0.040175221242 i$	$0.015840773241 + 0.040175221243 i$	$6.5 \cdot 10^{-12}$
1/3	above	$\emptyset$	$0.016581162594 - 0.039504714589 i$	$0.016581162601 - 0.039504714589 i$	$6.4 \cdot 10^{-12}$
1/4	below	$\emptyset$	$0.014455842876 + 0.039068895117 i$	$0.014455842881 + 0.039068895118 i$	$5.3 \cdot 10^{-12}$
1/4	above	$\emptyset$	$0.015760467204 - 0.037801851383 i$	$0.015760467209 - 0.037801851382 i$	$5.3 \cdot 10^{-12}$
2/5	below	$\emptyset$	$0.014434584508 + 0.037505178012 i$	$0.014434584513 + 0.037505178012 i$	$4.6 \cdot 10^{-12}$
2/5	above	$\emptyset$	$0.014296912442 - 0.037686119109 i$	$0.014296912446 - 0.037686119110 i$	$4.6 \cdot 10^{-12}$
3/7	above	$\emptyset$	$0.013301284920 - 0.037190044710 i$	$0.013301284861 - 0.037190044712 i$	$5.9 \cdot 10^{-11}$
3/7	below	$\emptyset$	$0.013998687363 + 0.036367863348 i$	$0.013998687363 + 0.036367863334 i$	$1.4 \cdot 10^{-11}$
1/3	above	2	$0.046849904480 - 0.018242741860 i$	$0.046849904482 - 0.018242741860 i$	$2.7 \cdot 10^{-12}$
1/3	below	2	$0.046510029333 + 0.018510459943 i$	$0.046510029336 + 0.018510459943 i$	$2.8 \cdot 10^{-12}$
1/3	above	3	$0.052913142048 - 0.001728643502 i$	$0.052913142050 - 0.001728643502 i$	$1.7 \cdot 10^{-12}$
1/2	below	3	$0.053307011907 + 0.002605955801 i$	$0.053307011909 + 0.002605955801 i$	$2.5 \cdot 10^{-12}$
1/3	below	3	$0.052713652883 + 0.001878998752 i$	$0.052713652885 + 0.001878998752 i$	$1.7 \cdot 10^{-12}$
1/1	below	3	$0.054529447219 + 0.005017189409 i$	$0.054529447226 + 0.005017189408 i$	$6.7 \cdot 10^{-12}$
1/3	below	(1, 2)	$0.053167372528 + 0.020247303769 i$	$0.053167372529 + 0.020247303768 i$	$1.6 \cdot 10^{-12}$

TABLE 2. One-sided limits of κ at rationals: the tail rule of Conjecture 6.1 against the bulbs $[0; a_1, \dots, a_\ell, N, \tau]$, extrapolated from $N = 128, \dots, 2048$ (degree four against degree three: at most $1.3 \cdot 10^{-9}$).

Other components. For a hyperbolic component W , call the first-return map of the parabolic cycle at its root the *root germ* of W . We conjecture (the root principle) that the limits of κ and of G along the satellites of W at angles $[0; N, \tau]$ and $1 - [0; N, \tau]$ are given by the rule above, with the upper and lower horn germs of the root germ. We tested this for the following components:

- the period-two disc: its root germ is $f_{e^{2\pi i/2}}$, and its constant $\kappa_{1/2}^+$ is known to 32 digits, and the best bulb fits agree with it to $2 \cdot 10^{-9}$ (uncertainty $2 \cdot 10^{-7}$);
- the period-three disc $W_{1/3}$: both sides follow $\mathcal{P}_{1/3}^\pm$, and the other three candidate germs miss by 10^{-3} or more;
- the real period-three component, whose root is a cusp with root germ normalized to $v + v^2 - \frac{2}{49}v^3 + \dots$ (exactly): agreement to $4 \cdot 10^{-10}$, with a kinematic first-order term;
- a period-six component at the $\frac{1}{3}$ -root of the period-two disc.

At that last root the root germ is not formally conjugate to $f_{e^{2\pi i/3}}$. Its root index differs by $6 \cdot 10^{-2}$, yet its horn constants differ from $\kappa_{1/3}^\pm$ by only about $1.5 \cdot 10^{-4}$, and the satellites follow its own constants.

7. DEGREE THREE AND OUTLOOK

For $z^3 + c$, near the cusp of the main component, the germ is $v + v^2 + \frac{1}{3}v^3$. Its horn map gives $\kappa_0^{(3)} = 0.1435992780952916 - 0.0303116478985025 i$, and $\kappa^{(3)}(1/q) + i/(2\pi q) \rightarrow \kappa_0^{(3)}$ like q^{-2} . The universal first-order term is unchanged and the constant is not. Here each bulb has two points of multiplier -1 , and both of their sizes are predicted. Integer-relation searches (PSLQ) found no closed form for κ_0 , $\kappa_0^{(3)}$ or $\kappa_{1/2}^+$ in a small basis.

Open problems.

- (1) Prove Conjectures 3.3 and 4.2 via continuity of near-parabolic renormalization.
- (2) Prove that the fixed point $\mathcal{G}_* = \mathcal{P}(\mathcal{G}_*)$ exists and describe its spectrum.
- (3) Prove Conjecture 6.1, and explain the jumps $\kappa_{p/q}^+ - \kappa_{p/q}^-$. They are nearly imaginary and depend mainly on q .
- (4) Relate κ_0 to the Écalle–Voronin invariants of $z + z^2$ [4, 14].

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